Question 1: What is a random variable in probability theory?

Ans: A random variable is a function that assigns a numerical value to each possible outcome of a random experiment. It is generally represented by a capital letter such as X, Y, or Z.

For example, when a coin is tossed twice, the possible outcomes are:

S= {HH, HT, TH, TT}

Let X be the number of heads obtained. Then:

X(HH)= 2, X(HT)= 1, X(TH)= 1, X(TT)= 0.

Thus, X is a random variable because its numerical value depends on the outcome of the experiment.

A random variable does not mean that the variable itself is random. Rather, its value is determined by the random outcome of an experiment.

Question 2: What are the types of random variables?

Ans: A random variable is a function that assigns a numerical value to each possible outcome of a random experiment. Based on the nature of the values it can take, random variables are mainly classified into two types: Discrete Random Variable and Continuous Random Variable.

a) Discrete Random Variable: A discrete random variable is a random variable that can take a finite or countably infinite number of distinct values.

Its possible values can be counted and are usually whole numbers.

Examples:

Number of heads obtained when a coin is tossed.

Number of students present in a classroom.

Number of defective items in a production batch.

Number of goals scored in a football match.

Example:

If a coin is tossed 3 times and X represents the number of heads, then:

X= 0, 1, 2, 3

These values are countable, so X is a discrete random variable.

b) Continuous Random Variable: A continuous random variable is a random variable that can take any value within a given interval or range.

Its possible values are not countable because there can be infinitely many values between two numbers.

Examples:

Height of a person.

Weight of a person.

Temperature.

Time required to complete a task.

Distance travelled by a vehicle.

Example:

If X represents the height of a person, it could be:

X= 165, 165.2, 165.25, 165.253,...

Therefore, height is a continuous random variable.

Thus, random variables are mainly classified into discrete and continuous random variables. Discrete random variables deal with countable outcomes, whereas continuous random variables deal with measurable quantities that can take any value within a range.

Question 3: Explain the difference between discrete and continuous distributions.
Ans: A probability distribution describes how the probabilities of different possible values of a random variable are distributed. Based on whether the random variable takes countable or uncountable values, probability distributions are mainly classified as discrete and continuous distributions.

a) Discrete Distribution: A discrete probability distribution is a probability distribution for a discrete random variable, which takes a finite or countably infinite number of distinct values.
Examples: Binomial distribution, Poisson distribution, and Bernoulli distribution.

b) Continuous Distribution: A continuous probability distribution is a probability distribution for a continuous random variable, which can take any value within a given interval.
Examples: Normal distribution, Uniform distribution, and Exponential distribution.

Difference Between Discrete and Continuous Distributions

Basis	**Discrete Distribution**	**Continuous Distribution**
Random variable	Takes countable values	Takes any value within a range
Nature of values	Separate and distinct	Infinitely many possible values
Probability	P(X= x) can be greater than 0	P(X= x)= 0
Function used	Probability Mass Function (PMF)	Probability Density Function (PDF)
Probability calculation	Probabilities are added	Probabilities are calculated using integration
Typical data	Counted data	Measured data
Example	Number of students in a class	Height of a student
Common distributions	Binomial, Poisson	Normal, Uniform, Exponential

Therefore, the main difference is that a discrete distribution deals with countable values, while a continuous distribution deals with values that can vary continuously within an interval. Discrete distributions use a PMF, whereas continuous distributions use a PDF to describe the probability distribution.

Question 4: What is a binomial distribution, and how is it used in probability?

Ans: A binomial distribution is a discrete probability distribution that describes the probability of getting a certain number of successes in a fixed number of independent trials, where each trial has only two possible outcomes: success or failure.
For example, if a coin is tossed 5 times, we can use a binomial distribution to find the probability of getting exactly 3 heads.

Conditions of Binomial Distribution
A random experiment follows a binomial distribution when:

a. There is a fixed number of trials, denoted by n.
b. Each trial has only two possible outcomes — success or failure.
c. The probability of success “p” remains constant for every trial.
d. The trials are independent of each other.
e. We are interested in the number of successes.

Binomial Probability Formula
The probability of getting exactly x successes in n trials is:

$$P(X = x) = \binom{n}{x} p^{x}(1 - p)^{n-x}$$

Where:
n = total number of trials
x = number of successes
p = probability of success
1-p = probability of failure
$\binom{n}{x}$ = number of ways to obtain x successes

Example
Suppose a fair coin is tossed 4 times. What is the probability of getting exactly 2 heads?
Here, n= 4, x= 2, p= 1/2
Therefore,

P(X= 2) = $\binom{4}{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{2}$ = 6 *(1/16) = 3/8

So, the probability of getting exactly 2 heads is:
3/ 8= 0.375

Uses of Binomial Distribution
Binomial distribution is used when we need to count the number of times an event occurs in repeated trials. For example:

- Finding the number of heads in repeated coin tosses.
- Finding the number of defective products in a sample.
- Finding the number of students who pass an examination.
- Finding the number of customers who respond to an advertisement.
- Finding the number of successful predictions made by a model.

Thus, the binomial distribution is used in probability to calculate the likelihood of obtaining a specific number of successes in a fixed number of independent trials, when each trial has two possible outcomes and a constant probability of success.

Question 5: What is the standard normal distribution, and why is it important?

Ans: The standard normal distribution is a special type of normal probability distribution with a mean of 0 and a standard deviation of 1. It is a continuous, symmetric, bell-shaped distribution.
It is usually denoted by:
Z~N(0, 1)
where:
Z = standard normal random variable
Mean =0
Standard deviation =1
Standardization (Z-score)
Any normally distributed value can be converted into a standard normal value using the Z-score:
Z=$(X - \mu)/\sigma$
where X is the observed value, μ is the mean, and σ is the standard deviation.
For example, if the mean marks are 60, the standard deviation is 10, and a student scores 80:
This means the student's score is 2 standard deviations above the mean.

It's Importance:
The standard normal distribution is important because:

a. It allows comparison: Values from different normal distributions can be compared using their Z-scores.
b. It helps calculate probabilities: Z-scores can be used with a standard normal table to find probabilities and areas under the curve.
c. It is widely used in statistics: It is used in hypothesis testing, confidence intervals, and statistical inference.
d. It simplifies calculations: Different normal distributions can be converted to the same standard normal distribution N(0, 1).
e. It is useful in data analysis: It helps identify how far an observation is from the average and can help detect unusual values.

Thus, the standard normal distribution is a normal distribution with mean 0 and standard deviation 1. Its main importance is that it provides a common scale through Z-scores, making probability calculations & comparisons easier in statistics and data analysis.

Question 6: What is the Central Limit Theorem (CLT), and why is it critical in statistics?

Ans: The Central Limit Theorem (CLT) is one of the most important concepts in probability and statistics. It states that when we take a sufficiently large number of independent random samples from a population, the distribution of the sample means tends to become approximately normal, regardless of the shape of the original population distribution.
In simple words, even if the original data is not normally distributed, the means of many sufficiently large samples will form a distribution that is approximately normal (bell-shaped).
Example

Suppose we have a population of people's incomes, and the income distribution is highly skewed.
If we repeatedly take random samples of, say, 30 people and calculate the mean income of each sample, then the distribution of these sample means will tend to look approximately like a normal distribution.

Why is CLT Critical in Statistics?
The Central Limit Theorem is important because:

a. It allows us to use the normal distribution even when the original population is not normally distributed.
b. It helps in estimating population parameters such as the population mean.
c. It forms the basis for confidence intervals and hypothesis testing.
d. It makes statistical analysis of large samples much easier.
e. It is widely used in data analysis, surveys, experiments, and machine learning.

Thus, the Central Limit Theorem explains why sample means tend to follow a normal distribution when the sample size is sufficiently large. It is critical because it provides the foundation for many statistical methods, especially confidence intervals and hypothesis testing, and allows us to make conclusions about a population using sample data.

Question 7: What is the significance of confidence intervals in statistical analysis?
Ans: A confidence interval (CI) is a range of values calculated from sample data that is used to estimate an unknown population parameter, such as the population mean or proportion, with a specified level of confidence.
Example
Suppose a survey estimates that the average income of a population is ₹30,000, and the 95% confidence interval is: $28000 \le \mu \le 32000$
This means the sample provides an estimated range of ₹28,000 to ₹32,000 for the population mean.
Significance of Confidence Intervals
Confidence intervals are important because:

1. They show uncertainty: They indicate how much uncertainty exists in our sample estimate.
2. They provide a range instead of a single value: A range gives more information than just a point estimate.
3. They help make statistical decisions: Confidence intervals are useful in hypothesis testing and decision-making.
4. They indicate precision: A narrower interval generally indicates a more precise estimate, while a wider interval indicates greater uncertainty.
5. They are useful for population estimation: They allow us to estimate population parameters using sample data.

The commonly used confidence levels are:
90%
95%
99%

A higher confidence level generally results in a wider confidence interval.

Thus, confidence intervals are significant in statistical analysis because they provide a range of plausible values for an unknown population parameter, while also showing the uncertainty and precision of the estimate. They are widely used in research, surveys, business analysis, and scientific studies.

Question 8: What is the concept of expected value in a probability distribution?

Ans: The expected value of a random variable is the weighted average of all its possible values, where each value is weighted according to its probability of occurring.

For a discrete random variable X, the expected value is calculated as:

$$E(X) = \sum xP(X = x)$$

where:

x = possible value of the random variable

P(X= x) = probability of that value

E(X) = expected value

Example

Suppose a game has the following possible outcomes:

Outcome (X)	Probability P(X)
10	0.2
20	0.5
30	0.3

Then: E(X)= 2+ 10+ 9 =21

Therefore, the expected value is 21.

This does not mean that the actual outcome must be 21. It means that if the experiment is repeated many times, the average outcome tends to approach 21.

Importance of Expected Value

Expected value is important because:

1. It provides the long-run average outcome of a random experiment.
2. It helps in decision-making under uncertainty.
3. It is widely used in risk analysis, finance, insurance, and business.
4. It forms the basis for other statistical concepts such as variance and standard deviation.

Question 9: Write a Python program to generate 1000 random numbers from a normal distribution with mean = 50 and standard deviation = 5. Compute its mean and standard deviation using NumPy, and draw a histogram to visualize the distribution.

Ans: The following Python program generates 1000 random numbers from a normal distribution with mean = 50 and standard deviation = 5. It then calculates the mean and standard deviation using NumPy and displays a histogram.

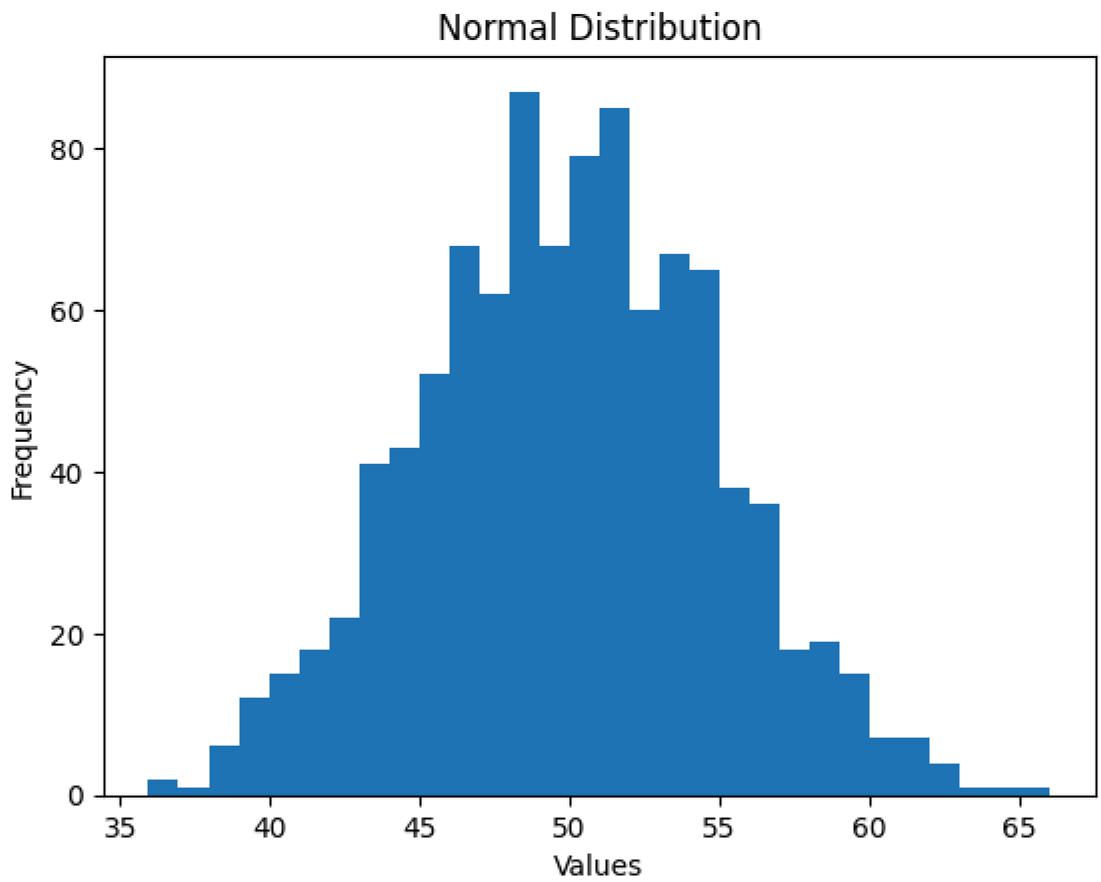
```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Generate 1000 random numbers
data = np.random.normal(loc=50, scale=5, size=1000)

# Calculate mean and standard deviation
mean = np.mean(data)
std = np.std(data)

# Display the results
print("Mean:", mean)
print("Standard Deviation:", std)

# Draw histogram
plt.hist(data, bins=30)
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.title("Normal Distribution")
plt.show()
```

Sample Output

Since the numbers are randomly generated, the output will be slightly different each time.

```
Mean: 49.966
Standard Deviation: 4.915
```



Question 10: You are working as a data analyst for a retail company. The company has collected daily sales data for 2 years and wants you to identify the overall sales trend.
daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
235, 260, 245, 250, 225, 270, 265, 255, 250, 260]

- **Explain how you would apply the Central Limit Theorem to estimate the average sales with a 95% confidence interval.**
- **Write the Python code to compute the mean sales and its confidence interval.**

Ans: The Central Limit Theorem (CLT) states that when the sample size is sufficiently large, the sampling distribution of the sample mean is approximately normally distributed, even if the original data is not normally distributed.

Here, we have daily sales data. We can use the sample mean and sample standard deviation to estimate the 95% confidence interval for the average daily sales.

Python Code

```
import numpy as np

# Daily sales data
daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
```

```
            235, 260, 245, 250, 225, 270, 265, 255, 250, 260]

# Convert to NumPy array
sales = np.array(daily_sales)

# Calculate sample size, mean and standard deviation
n = len(sales)
mean_sales = np.mean(sales)
std_sales = np.std(sales, ddof=1)

# Calculate standard error
standard_error = std_sales / np.sqrt(n)

# Calculate 95% confidence interval
margin_of_error = 1.96 * standard_error
lower = mean_sales - margin_of_error
upper = mean_sales + margin_of_error

# Display results
print("Mean Sales:", mean_sales)
print("Standard Deviation:", std_sales)
print("95% Confidence Interval:", (lower, upper))
```

Output

```
Mean Sales: 248.25
Standard Deviation: 17.265
95% Confidence Interval:(np.float64 (240.683),np.float64(255.817))
```